

Perovskite Photovoltaics in August 2026: Buried-Interface Chemistry, Defect Tolerance and the Certified Efficiency Frontier

Havid Aqoma^{1,2,3,4}

¹School of Energy and Chemical Engineering, Xiamen University Malaysia, Selangor Darul Ehsan 43900, Malaysia

²Kelip-kelip! Center of Excellence for Light Enabling Technologies, School of Energy and Chemical Engineering, Xiamen University Malaysia, Selangor Darul Ehsan 43900, Malaysia

³Centre of Excellence for Green and Advanced Technologies in Efficient Separations (GATES), School of Energy and Chemical Engineering, Xiamen University Malaysia, Selangor Darul Ehsan 43900, Malaysia

⁴Gulei Innovation Institute, Xiamen University, Zhangzhou 363200, China
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Abstract

Perovskite photovoltaics published 531 indexed works in August 2026, of which 144 were read closely and 82 are cited here. This review is organised mechanistically rather than chronologically, and separates what the month’s literature verified from what it claimed. The certified frontier advanced on three fronts, each through interfacial rather than bulk chemistry. A non-anchoring additive that suppresses self-assembled monolayer aggregation while leaving substrate binding sites available delivered a 33.66% certified perovskite/silicon tandem [1]; a thienyl self-assembled monolayer templating preferential crystallisation of the wide-bandgap subcell reached 32.56% certified on a tunnel-oxide passivated contact silicon bottom cell [2]; and a thiocyanate healing treatment driving secondary crystal growth into conformal columnar grains gave 30.77% certified on industrial textured silicon [3]. Single-junction certification clustered near 27.4%, led by redirection of a wet-interfacial iodide redox pathway at an acidic phosphonic-acid contact at 27.4% certified [4]. The largest certified area was a 61.2 cm² module at 17.35% certified, coated by a closed-loop slit-flushing antisolvent route in ambient air [5], which is also the month’s clearest statement of the area penalty. Only 18 of 144 closely read papers reported an independently certified power conversion efficiency at all. Operational-stability evidence was the weakest link: 5 papers reported a time to 80% of initial performance (T80), the longest at 1400 h [6], and 2 named a protocol from the International Summit on Organic Photovoltaic Stability (ISOS), so most lifetime claims this month cannot be compared against each other. Across the corpus an efficiency value was detected in at least 55.5% of abstracts, independent certification in at least 9.7%, and a complete open-circuit voltage, short-circuit current density and fill factor triplet in at least 4.1%. Three mechanistic conclusions follow. Buried-interface chemistry, not absorber composition, now sets the achievable open-circuit voltage. Defect tolerance has overtaken defect elimination as the practical design strategy. Halide homogenisation in wide-bandgap absorbers is the rate-limiting step for tandem integration, and it is the subcell least often optimised in the month’s work.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite;

tandem solar cells; operational stability

1. Introduction

Perovskite single junctions are now certified above 27% power conversion efficiency (PCE) and perovskite/silicon tandems above 33%, so the field’s open problems have moved from raw efficiency to whether a claimed efficiency is verifiable, whether it survives at module area, and whether it survives operation. In August 2026, 531 indexed works met the scope gate and 144 were read closely.

This review follows four mechanistic threads: buried-interface and contact chemistry, defect tolerance and ion migration, wide-bandgap absorbers for tandems, and operational stability, with scale-up as the constraint coupling them. Buried interfaces dictate band alignment and non-radiative recombination kinetics, while mobile ionic defects screen internal fields and accelerate chemical corrosion at selective contacts. In wide-bandgap absorbers required for tandems, light-induced halide segregation pins photovoltages, whereas rapid crystallization during large-area coating multiplies carrier-trapping boundaries across the absorber volume.

The month’s headline numbers are strong, the evidence behind them is unevenly reported, and this review separates the two.

2. The Certified Frontier in August 2026

Independent verification remains rare across current publications: only 18 of 144 closely read papers reported an independently certified power conversion efficiency (PCE). For two-terminal (2T) tandems, independent certification established a 33.66% ceiling for perovskite/silicon tandems alongside certified efficiencies of 32.56% and 30.77% [1,2,3], whereas all-perovskite tandem devices reached certified boundaries of 29.7% and 27.84% [7,8]. In single-junction positive-intrinsic-negative (p-i-n) devices, non-radiative recombination suppression defines a certified ceiling of 27.4%, closely corroborated by certified metrics of 27.2% and 26.77% [4,9,10].

A self-reported champion on a sub-0.1 cm² cell and a certified value on a module are different classes of evidence. Self-reported metrics run substantially ahead of certified baselines because uncertified testing on sub-0.1 cm² apertures artificially minimizes series resistance, avoids lateral coating non-uniformities, and omits interconnection dead space. Uncertified claims at larger areas illustrate this tension: self-reported efficiencies reached 27.7% on 25 cm² and 27.01% on 58.51 cm² [11,12], rivaling certified small-area cells. However, uncertified performance across similar apertures diverges steeply due to sheet-resistance and defect-density penalties, falling to 23.3% on 57.3 cm², 23.14% on 29.7 cm², and 17.35% on 61.2 cm² [5,13,14]. Without independent verification across macroscopic apertures, uncertified records mask interfacial heterogeneity and distort true progress.

3. Buried Interfaces and Contact Chemistry

Buried contacts in inverted (p-i-n) and regular (n-i-p) perovskite solar cells (PSCs) govern non-radiative recombination and charge extraction, yet unpassivated defects, band misalignment, and interfacial voiding compromise performance [15,16,17]. At metal-oxide electron-transport layers (ETLs), reactive surface species trigger mechanical voiding; gas-solid ethanol vapor reconstruction converts tin oxide (SnO₂) surface hydroxyls into anchored acetate and passivates oxygen vacancies for an uncertified 26.19% champion power conversion efficiency (PCE) [18]. Subliming volatile pyruvic acid at 145 °C similarly cleans SnO₂ to lower oxygen defects, delivering a 26.17% champion PCE

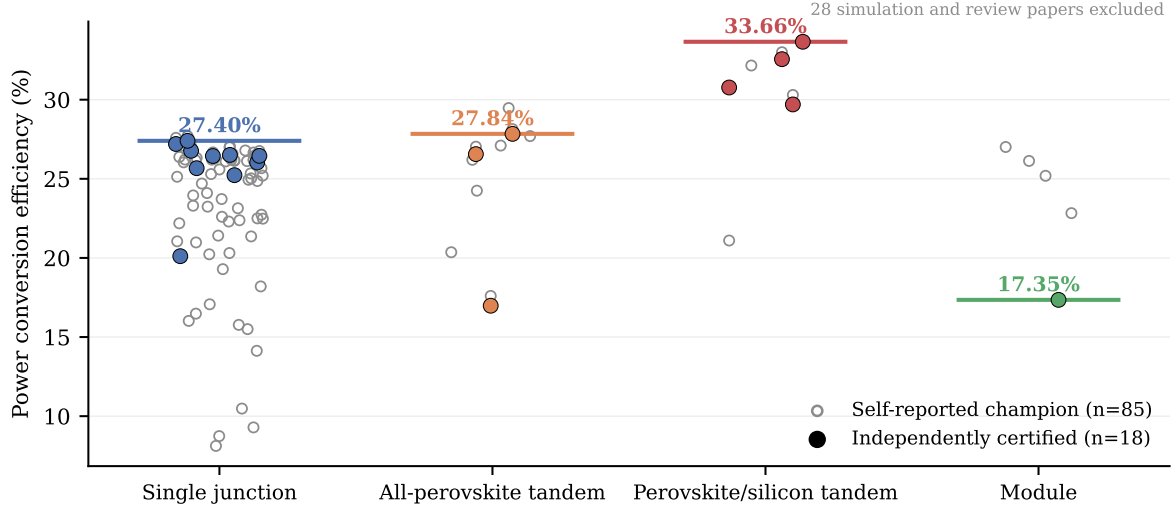


Figure 1. The certified frontier in August 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Horizontal bars mark the highest certified value in each family.

on a 1 cm² aperture with an 80% initial performance retention (T₈₀) of 900 hours [19]. Bidentate bridges such as 3,4-diaminobenzoic acid (DABA) bridge this boundary by anchoring carboxyl groups to Sn⁴⁺ while passivating undercoordinated lead (Pb²⁺) defects with amino groups [20].

At hole-transport layers (HTLs), interfacial acidity and defect states provoke analogous degradation. In methylammonium-free lead-tin devices, 4-(trifluoromethyl)benzamidinium hydrochloride (TFBA) suppresses acid-induced degradation by inhibiting sulfonate (-SO₃H) deprotonation on poly(styrenesulfonate) (PSS), securing an uncertified 22.48% champion PCE [21]. On nickel oxide (NiOx), ammonium thiocyanate (NH₄SCN) consumes surface hydroxyls and relieves lattice strain for a 21.41% champion PCE [22]. Decoupling self-assembled monolayer (SAM) chemisorption from bulk transport via a structural-compensation stack of sputtered NiOx and solution-processed hydroxyl-rich overlayers yields a 24.86% champion PCE across a 12.4 cm² aperture [23], while enamine-based SAM anchor engineering achieves 23.14% PCE and 83.11% fill factor (FF) on a 29.7 cm² mini-module [14].

However, SAM hole contacts frequently suffer from aggregation and dipole misalignment. Non-anchoring 1,3,5-tris(trifluoromethyl)benzene (TTMB) suppresses SAM aggregation while preserving substrate binding sites, reducing recombination and shunting to yield certified and stabilised 33.66% PCEs in perovskite/silicon tandems [1]. In wide-bandgap cells, building an interfacial hydrogen-bond network with glycine hydrochloride suppresses molecular clustering in [4-(3,6-dimethoxy-9H-carbazol-9-yl)butyl]phosphonic acid (MeO-4PACz) for a 22.38% champion PCE [24]. Asymmetric sulfur- and chlorine-functionalized molecules tune buried band offsets to achieve champion PCEs of 26.49% (25.70% steady-state) [25] and 26.38% [26], while co-assembling dipoles on NiOx yields a 20.36% four-terminal tandem champion PCE [27]. Yet dipolar engineering often lacks rigorous validation: bismuth oxychloride (BiOCl) flips interfacial dipoles alongside chlorine-iodine exchange to yield a certified 27.84% tandem PCE (champion 28.14%, stabilised 27.86%), but single-junction inverted devices reaching 27.25% (1.55 eV) and 20.24% (1.85 eV) PCE were restricted to an unscaled 0.09 cm² area [8], while dipole tuning with 2-fluorobenzene-1,3-dicarboxylic acid (m-FBC) reached

a 25.03% champion PCE whose 94% retention after 500 hours derived from unlabelled storage [28].

Buried contacts also suffer thermomechanical failure under operational stress. Anchoring [60]fullerene-phosphonic acid (C60-PA) on SnO₂ elevates maximum power point (MPP) tracked efficiency from an average of 18.6% to 20.0%, yet degrades under illuminated heating because fullerenes weaken interfacial adhesion [29]. Strengthening adhesion requires molecular templating: pairing precursor additives with phosphonic acid self-assembled monolayers (SAMs) enhances interfacial adhesion and suppresses groove depths for a 26.58% champion PCE [30], while self-assembled multilayers with [2-(10-(3,5-dimethoxyphenyl)-7H-benzo[c]carbazol-7-yl)ethyl]phosphonic acid (DMPA) secure certified 27.2% and champion 27.59% PCEs, retaining 94.5% efficiency after 1600 hours at 65 °C under 1 Sun illumination [9]. Coordinate bonding offers comparable reinforcement: thienyl-based additives template (111) crystallization and segregate at grain boundaries to form passivating lead-sulfur (Pb-S) bonds, yielding a certified 32.56% (champion 33.0%) tandem PCE [2]. Pb-S coordination via dibutyl sulfide induces a surface p-to-n transition for an uncertified 26.28% champion PCE [31], while multi-arm donor-acceptor dipole bridges coordinate Pb²⁺ via C=N and methoxy groups to relieve residual strain, yielding a 26.79% champion PCE [32].

4. Defect Tolerance, Passivation and Ion Migration

Undercoordinated lead (Pb²⁺) sites and halide vacancies drive non-radiative recombination and initiate photochemical degradation. Multidentate Lewis bases, chelating additives, and biopolymers suppress these trap states by coordinating Pb²⁺ and arresting formamidinium (FA⁺) deprotonation or iodide oxidation [33,34,35,36,37,38]. Coordinating Pb²⁺ with phenylthiourea (TPU) retards ion migration by suppressing iodine vacancies and volatile iodine loss, achieving an uncertified champion power conversion efficiency (PCE) of 26.13% [33]. Ammonium alginate stabilizes FA⁺ against deprotonation and passivates Pb²⁺, increasing the fill factor (FF) from 80.2% to 85.1% in an uncertified 26.12% cell [34]. Related treatments using sulfamethoxazole (SMX) (champion 21.05%, average 20.31%) [35], a sulfonium-amino acid additive (champion 22.5%) [36], multi-pyridyl spirobifluorenes (champion 20.98% versus 18.67% for control) [37], and 3-(trifluoromethyl)thiophenol (22.30% PCE) [38] similarly neutralize coordination defects, yet these studies examine uncertified small-area cells without establishing thermal bond durability.

Beyond surface coordination, mitigating ion migration requires addressing lattice mechanics, strain, and kinetic barriers within the perovskite absorber [39,40,41,42,43,44]. Stiffening the framework through solid-solution-induced lattice hardening via partial lanthanum substitution for Pb increases the Young’s modulus from 27.96 to 35.71 GPa, delivering a certified steady-state efficiency of 26.03% (champion 26.54%) [39]. Regulating crystallographic strain further restricts ion migration channels: shifting lattice strain from tensile to compressive with bimolecular sulfonates delivers an uncertified champion 26.21% PCE [40], dynamic strain homogenization via supramolecular networks in tin-lead (Sn-Pb) perovskite solar cells (PSCs) preserves 84.7% efficiency after 450 h of maximum power point tracking (MPPT) (champion 24.1%) [41], and doping with 0.8 mol% fluorinated methylammonium iodide (FMAI) maintains operational stability over 2000 h (champion 25.67%) [42]. Extended defect formation presents an associated thermodynamic dilemma: dimethylammonium (DMA⁺) phase engineering suppresses Ruddlesden-Popper antiphase boundary (RP-APB) defects in beta-(DMA,Cs)PbI₃ without nucleating the photo-inactive delta phase from edge-sharing [PbI₆]4-octahedra (champion 20.23%) [43]. Defect healing also follows an explicit kinetic hierarchy: ordered anion-cation modification (OACM) demonstrates that anionic vacancies must be repaired before

cation passivation to match bromide diffusion kinetics, enabling a 1.83 eV wide-bandgap (WBG) cell with an open-circuit voltage (Voc) of 1.33 V and 19.21% stabilized efficiency (champion 26.2%) [44].

At device contacts, dimensional heterostructures and buried molecular interlayers block defect propagation and mobile ion accumulation [45,46,47,48,49,50,51]. A tilted one-dimensional (1D) on three-dimensional (3D) heterointerface directed by a polyfluorinated ammonium iodide regulator preserves vertical transport while physically obstructing defect propagation and ion migration (champion 24.94% PCE) [45]. In two-dimensional (2D) capping layers, octylammonium bromide (OABr) concentration dictates phase shifts toward higher n -values (champion 19.29% PCE) [48], whereas chemical polishing removes unreacted formamidinium iodide (FAI) and residual lead iodide (PbI₂) to enforce vertical 2D orientation (champion 21.36% PCE) [47]. Solid-state halide exchange using bromide-rich nanocrystals passivates iodide vacancies and suppresses ion drift (uncertified preprint champion 22.19%) [46]. At buried electron-transport layer (ETL) interfaces, 2-formylbenzenesulfonic acid sodium salt (2-FAS) bridges self-assembled monolayers (SAM) to perovskite (champion 26.21% PCE, 86.15% FF) [49], acesulfame potassium (Ace-K) chelates Sn⁴⁺ and oxygen vacancies on tin oxide (SnO₂) to eliminate voids, retaining 92.4% efficiency after 1000 h of MPPT on an uncertified 0.0729 cm² cell (champion 26.17%) [50], and transient lead thiocyanate (Pb(SCN)₂) vapor eliminates residual PbI₂ traps (champion 25.2%) [51].

5. Wide-Bandgap Absorbers and Tandem Integration

Solution deposition of hole-selective self-assembled monolayers (SAMs) on textured silicon bottom cells suffers from inhomogeneous conformality, where uneven coverage of [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz) precipitates residual lead iodide (PbI₂) and causes interfacial shunting [7,52]. Cooperative self-assembled monolayer (SAM) strategies incorporating co-adsorbents with a methoxy terminal group and flexible alkyl chain improve substrate wettability and molecular packing on pyramids, lifting the uncertified power conversion efficiency (PCE) of two-terminal (2T) perovskite/silicon tandems with 1.68 eV wide-bandgap (WBG) absorbers to 32.16% [53]. To prevent degradation during atomic layer deposition (ALD), sacrificial coordination suppresses SAM aggregation via hydrogen bonding and shields the anchored monolayer, enhancing adhesion (ASTM D3359) to achieve an operational T90 of 1265 hours and a certified PCE of 26.56% (champion 27.03%) on a miniature 0.062 cm² aperture [54]. Inorganic hole-transport layer (HTL) stacks offer an alternative to vulnerable organophosphonates: pairing nickel oxide (NiO_x) with dye N719 achieves 96.4% PCE retention after one year of outdoor maximum power point tracking (MPPT) [55]. In polymeric HTLs, dilauroyl peroxide doping of poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) raises the water contact angle from 60.18° to 80.39°, boosting champion efficiency to 24.82% in n-i-p and 26.04% in p-i-n perovskite solar cells (PSCs) [56].

Concurrently, disordered nucleation, strain, and ion drift across textured pyramids compromise WBG absorber stability [57,58]. Pre-depositing a cesium chloride (CsCl) seed layer across uneven SAMs homogenizes organic precursor incorporation and eliminates interfacial PbI₂, securing a certified PCE of 29.7% (30.3% champion) [7]. Solvent-free sequential vapor evaporation avoids liquid dewetting entirely; in situ X-ray diffraction (XRD) confirms accelerated lead halide conversion on textured silicon over planar substrates, yielding a stabilised 27.0% PCE [59]. Post-deposition healing with methylammonium thiocyanate (MASCN) induces secondary crystallization into columnar grains spanning the full film thickness, enabling an encapsulated 1.164 cm² device to achieve a certified 30.77% PCE and retain initial performance over 3400 hours of continuous one-sun MPPT in ambient conditions

[3]. To counteract mechanical distortion, two-dimensional (2D) perovskite seeding directs regular compressive strain into the three-dimensional (3D) WBG lattice, delivering 20.83% PCE with an 84.72% fill factor (FF) in single junctions and 29.47% in tandems [60]. In the bulk absorber, ion modulation elevates the ion migration activation barrier, lifting uncertified champion tandem PCE to 33.02% from 31.12% for control devices [61].

At the tandem interconnect, optical and resistive losses restrict device operation. In perovskite-organic tandems, a zinc sulfide (ZnS) interlayer converts isolated gold clusters into laterally planarized domains, establishing low-loss recombination pathways at only $\sim 77.4\%$ metallic coverage while suppressing localized surface plasmon resonance to yield a 27.1% champion PCE and a T80 of 1400 hours [6]. Translating 2T tandems beyond 33%-34% laboratory benchmarks toward commercial modules requires overcoming texturing shunts, near-infrared (NIR) parasitic absorption, balance-of-system (BOS) costs, and the defect cascades that penalize large-area scaling [57,58].

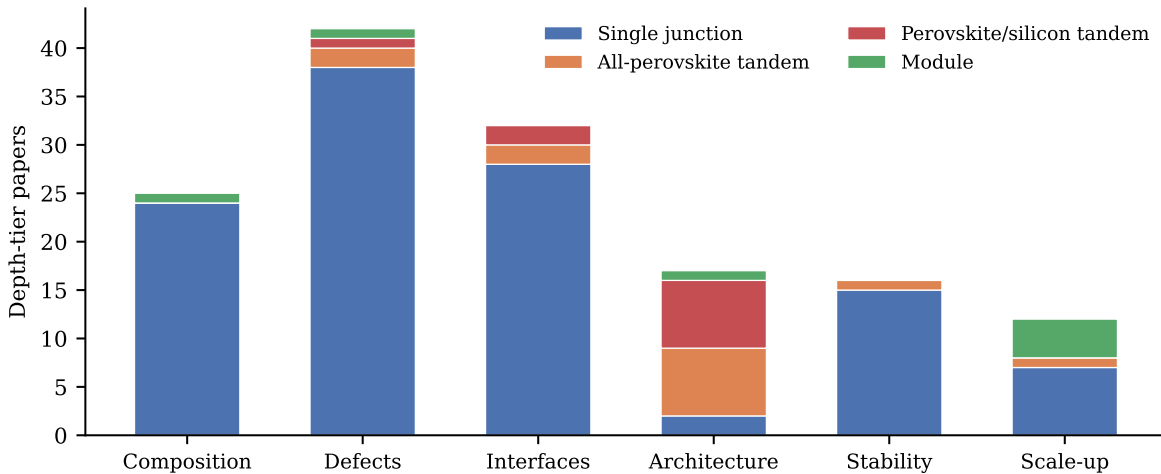


Figure 2. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

6. Operational Stability and Degradation Pathways

Operational degradation in inverted perovskite solar cells (PSCs) originates primarily from coupled interfacial redox reactions, lattice strain transitions, and electrode interdiffusion. At the buried hole-transport layer (HTL), acidic carbazole-based phosphonic acid self-assembled monolayers (SAMs) activate residual dimethyl sulfoxide (DMSO), driving iodide redox decomposition during wet processing [4]. Under subsequent light operation, this interfacial degradation cascades into the electron-transport layer (ETL): formamidinium iodide (FAI) photo-oxidation generates iodine radicals that induce phenyl-C61-butyric acid methyl ester (PCBM) dimerization, while fullerene species accelerate FAI deprotonation in a self-reinforcing photochemical loop [62]. Interfacial radical scavengers [62], cross-linkable self-assembled monolayer (SAM) linker architectures [63], and dipole-balancing molecular co-assemblies [64,65,66] mitigate these decomposition cascades. Nevertheless, stark discrepancies between uncertified champion power conversion efficiency (PCE) values and certified metrics, including a champion 27.06% PCE dropping to a certified 20.11% on small 0.09 cm² apertures [66], reveal unresolved interfacial instability and measurement drift.

Within the perovskite absorber, vertical lattice heterogeneity accelerates lead and iodide vacancy for-

mation under operating stress [67]. Thermal and operational biases exacerbate internal lattice strain, promoting polymorphic degradation from the photoactive α phase to the non-photoactive δ phase; relieving this strain through interstitial incorporation of 2 mol% sodium [68] or templating via two-dimensional (2D) intermediate phases [69] stabilizes the active crystal lattice. At the rear contact, halide migration triggers severe metal interdiffusion, requiring diffusion barriers such as 5 nm chromium interlayers to prevent contact failure [70,71].

A critical systemic weakness across these investigations is the severe lack of standardized aging protocols. Only 5 of 144 surveyed papers reported a time to 80% initial performance (T80) value: 1400 h [6], 900 h [19], 864 h [71], 800 h [68], and 720 h [72]. Concurrently, only 2 studies adhered to formal International Summit on Organic Photovoltaic Stability (ISOS) protocols, specifically ISOS-L-1 [64] and ISOS-L-2 [62]. Most stability tests rely on unencapsulated continuous illumination without standardized maximum power point tracking (MPPT) [73]. Consequently, existing literature primarily probes short-term photochemical burn-in, obscuring critical field degradation pathways including reverse-bias breakdown, moisture-accelerated ion migration, and thermo-mechanical fatigue under diurnal operational cycling.

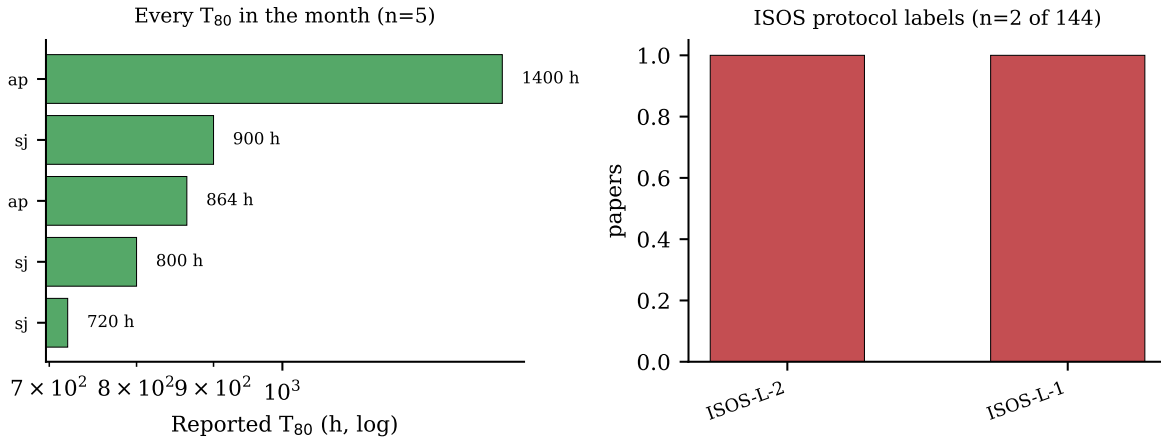


Figure 3. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

7. Scale-Up and the Area Penalty

The scaling of perovskite solar cells (PSCs) from laboratory coupons to large-area modules incurs an efficiency penalty dominated by spatial mass-transfer disparities, stochastic nucleation, and resistive or environmental losses. Non-uniform drying across decimeter areas induces pinholes and phase segregation; replacing vertical diffusion-limited solvent evaporation with convective or engineered antisolvent extraction suppresses these defects [5,11,74]. Closed-loop slit-flushing antisolvent deposition imposes horizontal forced convection over 100×100 mm substrates to achieve a certified power conversion efficiency (PCE) of 17.35% on a 61.2 cm² module [5]. Static antisolvent bathing with isopropanol admixtures homogenizes crystallization into larger crystalline nanodomains verified by grazing-incidence wide-angle X-ray scattering (GIWAXS), enabling an uncertified 27.7% champion PCE on a 25 cm² two-terminal (2T) tandem device [11], while polar additives in slot-die precursors yield 30 nm films with 1.86 nm roughness over 100 cm² [74]. In parallel, chemical coordination and heterogeneous seeding regulate nucleation barriers to prevent premature precipitation [12,75,76].

Polyfluoroarene additives elevate effective nucleation barriers to convert rapid aggregation into uniform film growth, achieving an uncertified module champion PCE of 27.01% on 58.51 cm² with small-area non-radiative voltage losses of 55 mV [12]. Purine tautomers alter lead-nitrogen coordination and hydrogen bonding, confirmed by density functional theory (DFT), yielding a certified 26.51% PCE (champion 27.05%) at 12.95 cm² [75], whereas cesium lead triiodide (CsPbI₃) nanocube seeding enforces controlled crystallization and depth-dependent Cs⁺ incorporation in slot-die coated formamidinium lead triiodide (FAPbI₃) [76]. Beyond nucleation, solid-state transformation barriers and resistive contacts limit module performance [77,78]. Laser-structured channels in lead iodide (PbI₂) provide deterministic diffusion pathways that lower phase transformation activation barriers, reaching an uncertified 22.83% PCE across 22.95 cm² modules [77]. For charge extraction, cerium iodide (CeI₃) interfacial modifiers establish inorganic hole contacts, yielding 25.19% PCE in 0.113 cm² cells and 23.37% in 21 cm² modules with a 98.3% geometric fill factor (GFF) [78]. Finally, scalable slot-die processing remains constrained by environmental sensitivity and mechanical degradation [79,80]. Ambient humidity induces marked PCE degradation when exceeding 30% relative humidity (RH) [79], yet incorporating Ti₃C₂T_x MXene nanospheres extends continuous illumination stability from 400 h to over 1700 h in all-slot-die-coated PSCs, albeit at an uncertified 18.2% champion PCE [80].

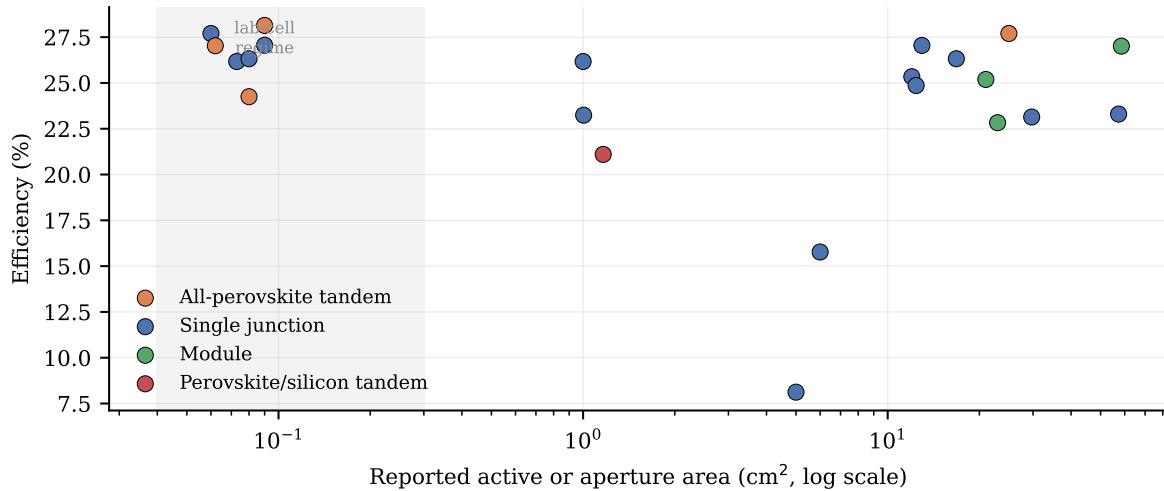


Figure 4. Efficiency against reported active or aperture area. The shaded band marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

8. Research Gaps and Outlook

The performance divergence between single junctions and tandems defines the first gap: the best certified perovskite/silicon tandem this month was 33.66% power conversion efficiency (PCE) [1], while certified single junctions clustered near 27%, showing that the wide-bandgap (WBG) subcell limiting the tandem is not the one most papers optimise. While chemical strategies claim to homogenize mixed halides in pristine films [81,82], whether halide homogenisation in wide-bandgap absorbers suppresses phase segregation under operating bias remains unproven because photocarrier gradients trigger iodide clustering. Operando spatially resolved photoluminescence (PL) spectroscopy under continuous maximum power point tracking (MPPT) load at elevated temperatures would resolve whether compositional uniformity survives operational electric fields.

A second gap concerns buried-interface chemistry that survives thermal cycling rather than only lifting initial efficiency. Molecules and self-assembled monolayers (SAMs) passivate buried defects to lift open-circuit voltage (Voc) on pristine substrates [63,66], but mismatched thermal expansion coefficients generate interfacial shear strain that induces delamination. Only 5 of 144 closely read papers reported a time to 80% of initial performance (T80) lifetime, the longest 1400 h [6], short against a 25-year module warranty. In situ temperature-dependent adhesion testing with impedance spectroscopy during thermal cycling would reveal whether interfacial bonding survives thermomechanical stress.

A third gap lies in ion-migration timescales measured under load rather than in the dark. Passivation schemes report reduced ion drift [44,61], yet extracting activation energies in the dark ignores how photogenerated carriers screen internal fields and accelerate vacancy migration under illumination. Operando transient ion-drift measurements under illumination during continuous MPPT would resolve ionic transit timescales and defect accumulation under load.

The fourth gap is whether passivation that lifts Voc on a 0.1 cm² cell survives slot-die coating at module area. Only 27 of 531 works addressed scale-up at all, and the largest reported aperture was 61.2 cm² [5], orders of magnitude below a commercial module. Additives that distribute evenly during spin coating aggregate under meniscus shear, creating non-uniform trap landscapes [12,76]. Operando PL imaging during slot-die coating would track drying kinetics and isolate spatial Voc losses. Standardizing stability assessments to International Summit on Organic Photovoltaic Stability (ISOS) protocols under operational load and validating passivation across large-area deposition would make next month’s literature considerably more useful.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the month, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. Abstract text and full texts are not redistributed.

AI Usage Declaration

This manuscript was prepared as part of our Research Project for making a reliable Autonomous Researcher Agent. The agent executed the monthly review pipeline without human intervention: it harvested the indexed literature for the month through the OpenAlex, Semantic Scholar, Crossref and arXiv interfaces, applied a deterministic topical filter, ranked and selected the papers read closely, extracted every quantitative claim into a structured record binding each number to a verbatim quotation from that paper’s own abstract, computed the statistics and figures from those records, and drafted this text from the resulting evidence briefs. Literature harvesting, selection,

statistics, figures and gate checking are deterministic scripts rather than model output. Structured extraction used qwen3.8-flash through the opencode interface; the prose was written by gemini-3.8-flash-high through the agy interface; independent review used opus at high reasoning effort. No figure is a model-generated image: every panel is plotted directly from the extracted records. Every quantitative statement in the main text resolves to the machine-generated statistics file shipped as Supplementary Information, and every citation resolves to a record carrying its source quotation. The author reviewed and edited the content, verified the selection, and takes full responsibility for the content of the published article.

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